Interchanging the expressions on both sides of an equation, does not make any change in the equation. For example, $x+3=5$ and $5=x+3$ both are same.

But, if the expressions on both sides of an inequation are interchanged, the sign of inequality must be reversed.

For example, $30 > 20$ is the same as $20 < 30$ and $-18 < -9$ is the same as $-9 > -18$.

Example 3.10 Solve: $2x + 4 < 18$, where x is a natural number.

F4Y6Z3

Solution

$$2x + 4 < 18$$

$$2x + 4 - 4 < 18 - 4 \quad \text{[Subtracting 4 from both sides]}$$

$$2x < 14 \quad \text{[Divide by 2 on both sides]}$$

$$x < 7$$

Since the solution belongs to natural numbers, that are less than 7, we take the values of the x as 1, 2, 3, 4, 5 and 6.

Therefore, the solutions are 1, 2, 3, 4, 5 and 6.

Example 3.11 Solve: $5 - 7x \geq 33$, where x is an integer.

Solution

$$5 - 7x \geq 33$$

$$5 - 5 - 7x \geq 33 - 5 \quad \text{[Subtracting 5 from both sides]}$$

$$-7x \geq 28$$

$$\frac{-7x}{-7} \geq \frac{28}{-7} \quad \text{[Dividing both sides by } -7\text{]}$$

$x \leq -4$ [since, it is divided by a negative number, the inequality is reversed]

Since, solution belongs to the set of integers, that are less than -4, we take the values of x as $-4, -5, -6, \ldots$

Therefore, the solutions are $-4, -5, -6, \ldots$.

Think

Hameed saw a stranger in the street. He told his parent, "The stranger's age is between 40 to 45 years, and his height is between 160 to 170 *cm*".

Convert the above verbal statement into an algebraic inequations by using x and y as variables of age and height.

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